

## Project 6 – Footprinting, Reconnaissance, and Scanning

### Answer Sheet

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### Part 1 – Passive Reconnaissance / OSINT

#### Task 1 – Whois: Domain Identity & Ownership

Instructions: Use a public Whois lookup for testphp.vulnweb.com (or its parent domain) and record the requested information below.

Domain:

vulnweb.com

Registrar:

Gandi SAS

Organization / Owner:

Invicti Security Limited

Creation Date:

2010-06-14

Expiration Date:

2027-06-14

Name Servers:

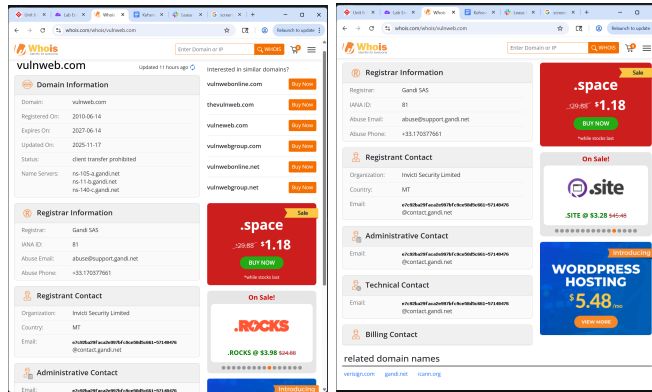
ns-105-a.gandi.net

ns-11-b.gandi.net

ns-140-c.gandi.net

Notes: Domain is [vulnweb.com](https://vulnweb.com) and the registrar is Gandi SAS. The organization and owner of the website is Invicti Security Limited. Created on 2010-06-14, and expiration date 2027-06-14.

## Screenshots of Whoislookup



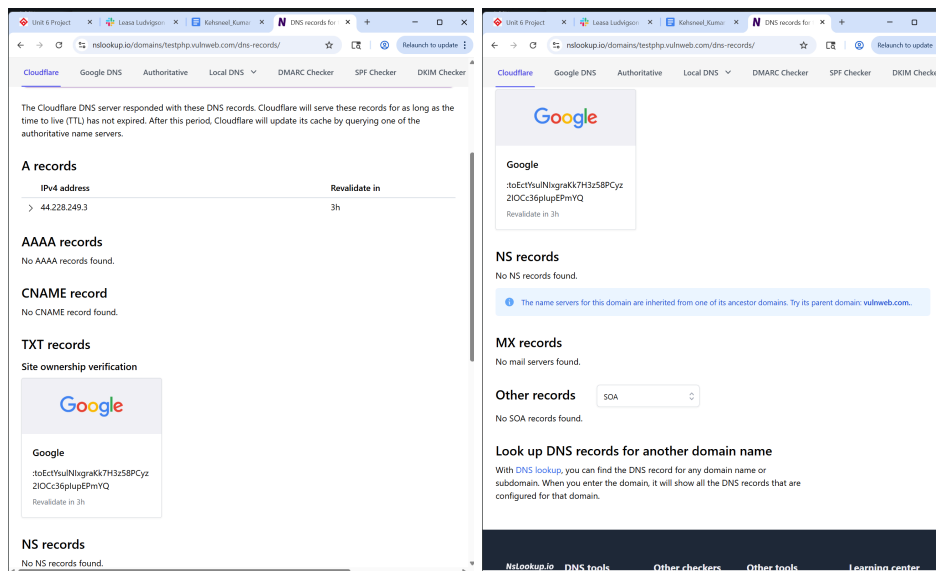
## Task 2 – DNS Basics

Instructions: Use nslookup or dig for testphp.vulnweb.com and summarize the DNS information.

Command used: <https://www.nslookup.io/domains/testphp.vulnweb.com/dns-records/>

IP address(es) observed: 44.228.249.3

Record type(s) (A / CNAME / TXT, etc.): AAAA Records, CNAME Records, TXT Records, NS Records.



Additional notes: For [testphp.vulnweb.com](http://testphp.vulnweb.com) I observed the ipv4 address of 44.228.249.3. There were not any notable record types found.

### Task 3 – Hosting & Technology Stack

Instructions: Use Netcraft Site Report or a similar public site-technology report for testphp.vulnweb.com.

Tool used: Net craft <https://sitereport.netcraft.com/?url=http://testphp.vulnweb.com>

Hosting provider / ASN: Amazon - US West (Oregon) datacenter, the ASN [AS16509](#).

Country / Region: United States

Web server / tech stack reported: Web server/ tech stack was not reported.

Other observations: It was hosted in the United States. Web server/ tech stack was not reported. The site only runs on HTTP No SSL/TLS encryption which I found very interesting.

### Screenshots for Tasks 3

The image displays two side-by-side screenshots of the Netcraft website report for the URL <http://testphp.vulnweb.com>.

**Left Screenshot (Background and Network sections):**

- Background:** Site title is "http://testphp.vulnweb.com". Site rank is 1439. Description is "Not Present".
- Network:** Site is "http://testphp.vulnweb.com". Netblock Owner is "Amazon.com, Inc.". Hosting company is "Amazon - US West (Oregon) datacenter". Hosting country is "US". IPv4 address is "44.228.249.3".

**Right Screenshot (IP delegation and SSL/TLS sections):**

- IP delegation:** IPv4 address (44.228.249.3) is listed with a table of IP ranges and their descriptions.
- SSL/TLS:** This is not a HTTPS site. If you're looking for SSL/TLS information try the [HTTPS site report](#).

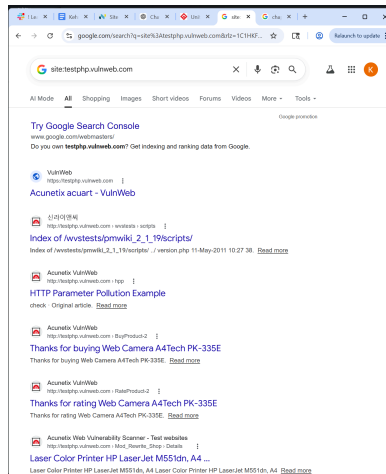
| IP range                  | Country       | Name                     | Description                            |
|---------------------------|---------------|--------------------------|----------------------------------------|
| ::ffff:0:0:0:0/96         | United States | IANA-IPv4 MAPPED ADDRESS | Internet Assigned Numbers Authority    |
| 44.0.0.0-44.255.255.255   | United States | NET44                    | American Registry for Internet Numbers |
| 44.192.0.0-44.255.255.255 | United States | AMAZON-4                 | Amazon.com, Inc.                       |
| 44.224.0.0-44.255.255.255 | United States | AMAZON-2PDK              | Amazon.com, Inc.                       |
| 44.228.249.3              | United States | AMAZON-2PDK              | Amazon.com, Inc.                       |

#### Task 4 – Google Dorks on testphp.vulnweb.com

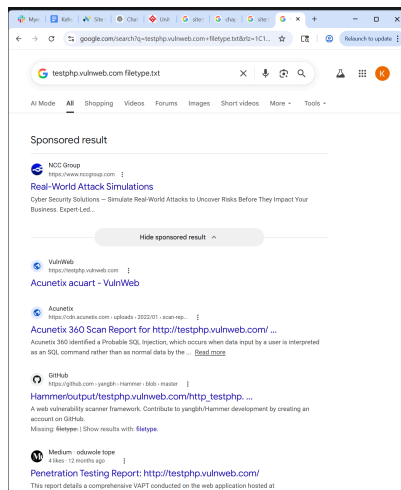
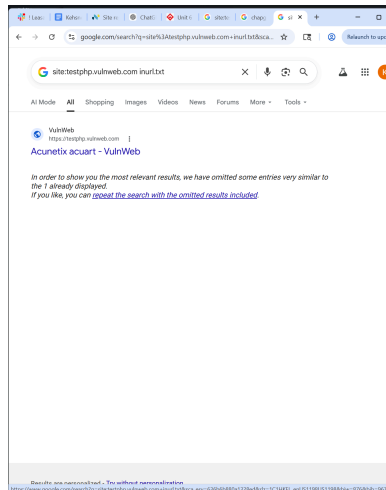
Instructions: Run at least three Google queries using operators (site:, inurl:, filetype:, "quoted phrase"). For each query, record one interesting result and why it might matter.

| Query                                                                | Interesting Result<br>(Title/URL)                                               | Why this might matter<br>(attacker or defender)                                                                                                                                                                                                                  |
|----------------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.site: <a href="http://testphp.vulnweb.com">testphp.vulnweb.com</a> | 1. <a href="#">Thanks for buying Web Camera A4Tech PK-335E</a>                  | 1. An attacker or defender would care about that message because it could indicate a successful purchase transaction, which might be used to verify user activity or detect phishing, fraud, or unauthorized orders.                                             |
| 2.site:testphp.vulnweb.com<br>inurl.txt                              | 2. <a href="#">Acunetix acuart - VulnWeb</a>                                    | 2.An attacker or defender would care about <b>Acuart</b> on <b>Acunetix's</b> VulnWeb because it's a vulnerable application that exposes common web security flaws, making it useful for practicing exploitation or testing detection and mitigation techniques. |
| 3.testphp.vulnweb.com<br>filetype.txt                                | 3. <a href="#">Acunetix 360 Scan Report for http://testphp.vulnweb.com/ ...</a> | 3.An attacker or defender would care about an <b>Acunetix 360 scan report</b> for <a href="http://testphp.vulnweb.com">testphp.vulnweb.com</a> because it details discovered vulnerabilities, which could be exploited if misused.                               |

## Screenshots for task 4



Screenshot filenames (if any):



## Task 5 – Light OSINT on fullstackacademy.com (Google Only)

Instructions: Run at least three Google searches using site:fullstackacademy.com or its subdomains with at least one other operator. Record an example result and why it could matter from an OSINT or security perspective.

| Query                                                                                   | Example Result (Title/URL)                                                          | Why this might matter                                                                                                       |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| 1.site: <a href="https://fullstackacademy.com">fullstackacademy.com</a><br>"curriculum" | 1. <a href="https://fullstackacademy.com/USD-Cyber-Syllabus">USD Cyber Syllabus</a> | 1. An attacker or defender would care about<br><a href="https://fullstackacademy.com/USD-Cyber-Syllabus">site:fullstack</a> |

2.site:fullstackacademy.com  
filetype:pdf

3.site:fullstackacademy.com  
inurl:blog

2.[Eligibility Criteria](#)

3.[About Tech  
Bootcamps](#)

academy.com  
"curriculum"

because it helps  
quickly locate  
publicly exposed  
curriculum content.

2. An attacker or  
defender would  
care about

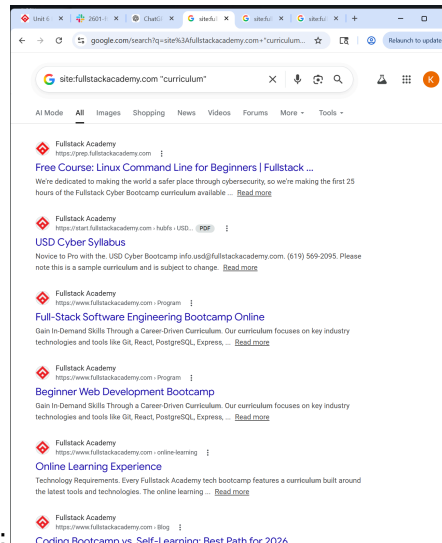
site:testphp.v  
ulnweb.com  
filetype:pdf

because it can  
reveal publicly  
accessible PDF  
documents that  
might contain  
sensitive  
information

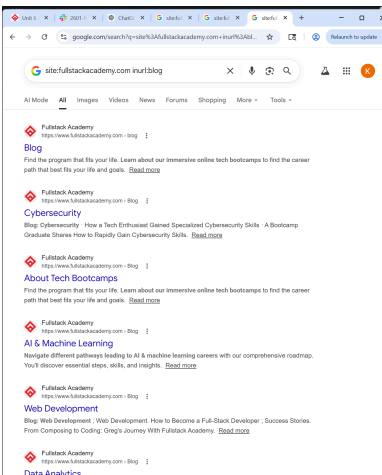
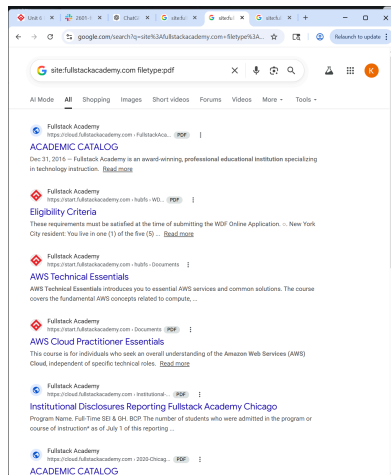
3. An attacker or  
defender would  
care about

site:fullstack  
academy.com  
inurl:blog

because it surfaces  
blog pages that may  
reveal technologies



Screenshot filenames (if any):



## Task 6 – Passive Recon Summary

Instructions: Summarize the overall exposure and key findings from your passive recon.

Summary for testphp.vulnweb.com (1–2 sentences): [Testphp.vulnweb.com](https://testphp.vulnweb.com) is a vulnerable demo website that appears to be highly exposed.

Summary for fullstackacademy.com (1–2 sentences): Fullstack academy is a website that is moderately exposed. Legitimate educational information without obvious exposure to anything serious.

Top 3–5 Passive Findings: Documents contain sensitive information useful for reconnaissance. Easily discoverable, and PDF files are accessible via the internet. Blogs pages are easily found on search engines such as google.com.

| Finding (what you discovered)                                                               | Tool / Query used                                  | Why it matters to a defender                                               |
|---------------------------------------------------------------------------------------------|----------------------------------------------------|----------------------------------------------------------------------------|
| 1.testphp.vulnweb, search results revealed pages with links.                                | 1. Google Dork<br>2. Google dork<br>3. Google Dork | 1. Parameterized endpoints can signal potential input vectors for attacks. |
| 2.testphp.vulnweb, PDF documents listed and are downloaded.                                 |                                                    | 2. Files may expose sensitive data or useful metadata.                     |
| 3. fullstackacademy.com website pages are easily found on search engines such as google.com |                                                    | 3. Blog posts may reveal technologies or internal practices.               |

Recommended follow-up action for a defender (1–2 sentences):

Follow up actions for a defender to audit all publicly available files for sensitive files and remove. Try regularly performing passive renaissance on domains such as google and OSINT tools..

## Part 2 – Active Reconnaissance / Scanning (Metasploitable Lab)

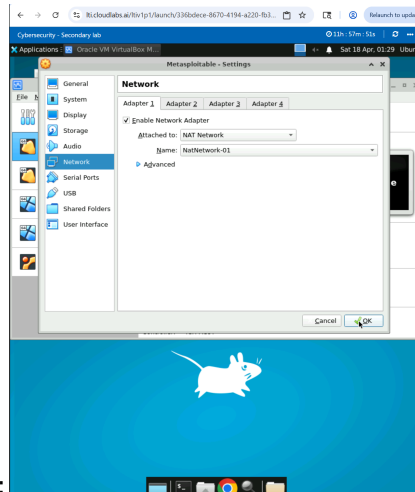
### Task 7 – Confirm Connectivity (Kali → Metasploitable)

Instructions: Ensure your Kali system can reach the Metasploitable VM and record the details below.

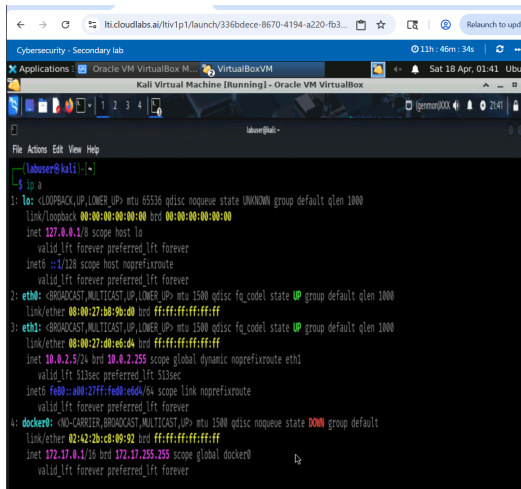
Metasploitable IP address:10.0.2.5/24

Ping command used:ip a

Result summary (e.g., number of successful replies): 4 Successful replies.



Screenshot filename(s):

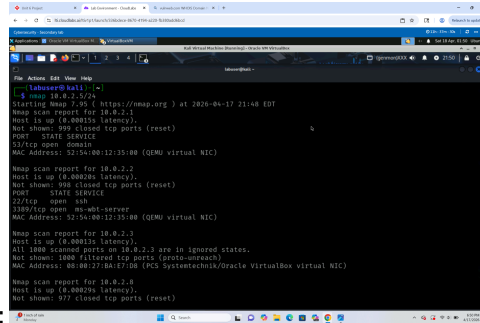


## Task 8 – Basic Nmap Scan of Metasploitable

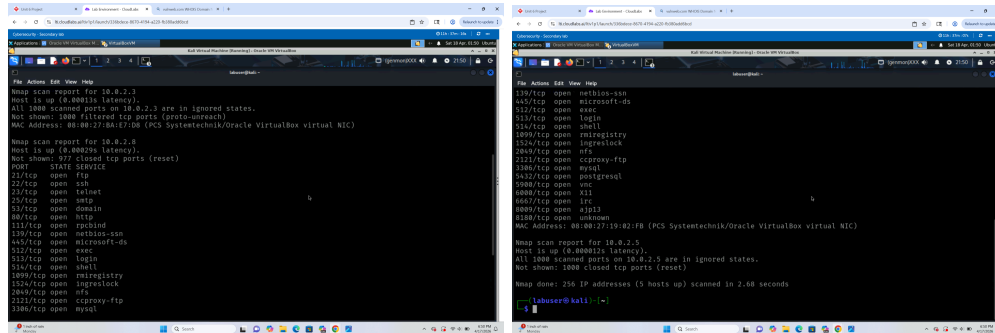
Instructions: Run a basic nmap scan against the Metasploitable IP address.

Nmap command used: `nmap 10.0.2.5/24`

General observations about the scan (number of open ports, anything that stands out): All 1000 scanned ports on 10.0.2.5 are in ignored states.



Screenshot filename(s):



## Task 9 – Port & Service Interpretation

Instructions: Choose at least five open ports from your nmap results. For each, describe the service, a potential risk, and a simple defense idea.

| Port / Protocol  | Service     | Version (if shown) | Example Risk           | Simple Defense     |
|------------------|-------------|--------------------|------------------------|--------------------|
| 1. 23/tcp open   | Telnet      |                    | 1. Remote comma        | 1. Disable telnet. |
| 2. 25/tcp open   | Smtp        |                    | nd- line               | 2. Needs proper    |
| 3. 53/tcp open   | Domain      |                    | access                 | configu            |
| 4. 80/tcp open   | Http        |                    | 2. Used for            | ration.            |
| 5. 1099/tcp open | rmiregistry |                    | sending email          | 3. Should restrict |
|                  |             |                    | betwee n               | zone transfer      |
|                  |             |                    | servers.               | s.                 |
|                  |             |                    | Targete d for          | 4. NONE            |
|                  |             |                    | spam emails            | 5. Should not be   |
|                  |             |                    | 3. Handles domain name | publicly accessi   |
|                  |             |                    |                        | ble unless         |

- |    |                                                    |                             |
|----|----------------------------------------------------|-----------------------------|
|    | resoluti<br>ons.                                   | strictly<br>controll<br>ed. |
| 4. | Standar<br>d web<br>server.                        |                             |
| 5. | Can be<br>risky if<br>expose<br>d<br>publicly<br>. |                             |

#### Task 10 – Active Recon Summary

Instructions: In 5–7 sentences, summarize what your scan results say about the Metasploitable system's exposure and what a defender should prioritize.

Summary: The scan of the Metasploitable system shows several exposed services, including Telnet (23), SMTP (25), DNS (53), HTTP (80), and Java RMI (1099), indicating a very large attack surface. In particular, Telnet and RMI are outdated and high-risk services that could allow unauthorized access or remote code execution. SMTP and DNS may also be abused if not properly configured, such as for spam relaying or DNS attacks. HTTP suggests a web service that could contain vulnerable applications commonly targeted in testing environments. A defender should prioritize disabling insecure services like Telnet, restricting or securing RMI, and properly hardening SMTP, DNS, and the web server to reduce exposure.